

**EX NO:1**

## **BUTTERWORTH IIR FILTERS**

### **1.1:Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.**

#### **Program (Matlab Code)**

```
clc; clear

Fs=10000; Fp=1000; Fst=1500;

[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);

[b,a]=butter(N,Wn);

freqz(b,a)

figure

% Magnitude & Phase

[h,n]=impz(b,a,50);

...title('Impulse Response')

figure

stepz(b,a)

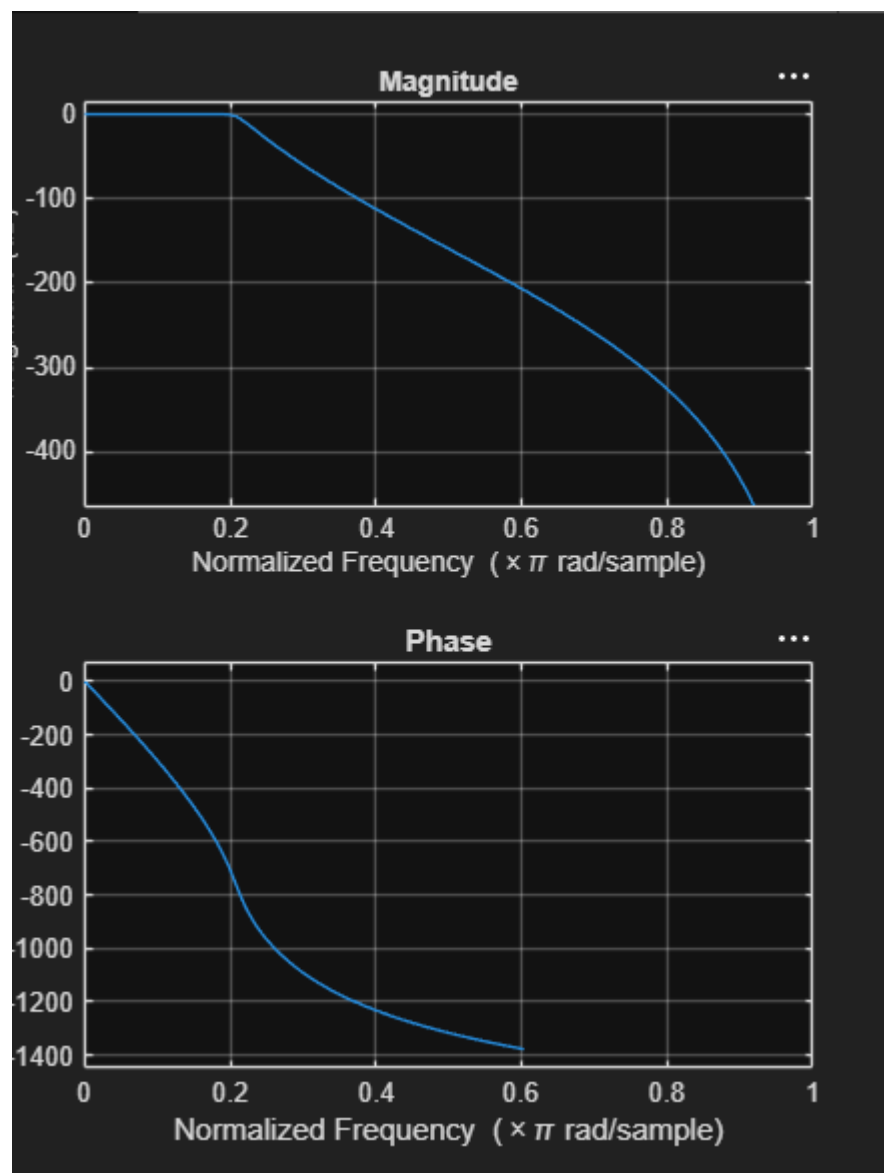
figure

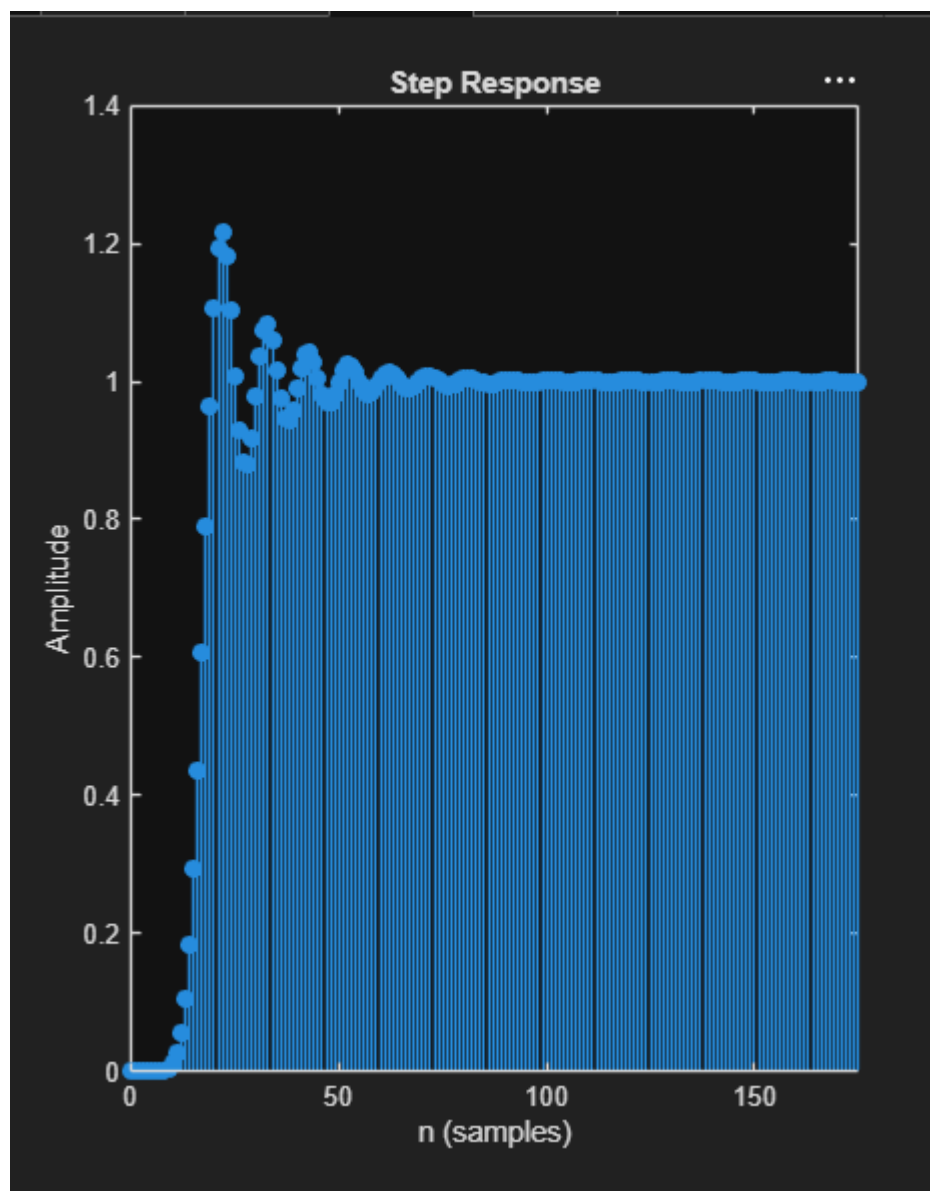
zplane(b,a)

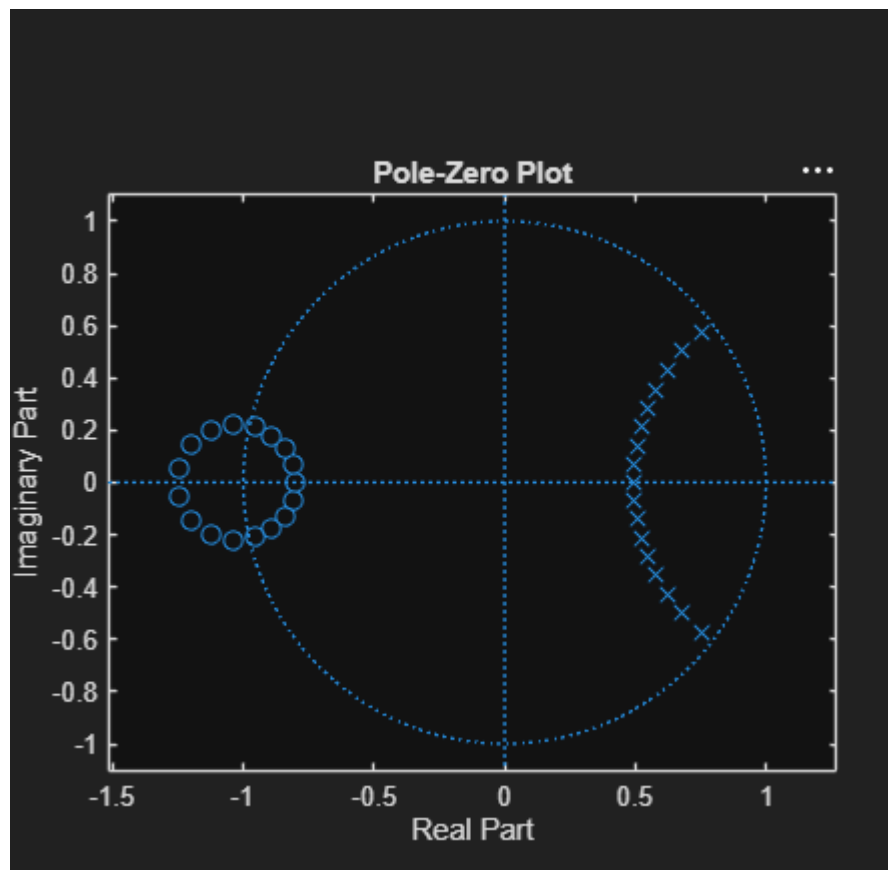
fprintf('Minimum Filter Order (N) = %d\n',N);

fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

**Output:**







Minimum Filter Order (N) = 17  
Cutoff Frequency (Wn) = 0.2083  
>>

## 1.2: Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

### Program (Matlab Code)

```
clc; clear

Fs=10000; Fp=3000; Fst=2000;

[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);

[b,a]=butter(N,Wn,'high');

% Magnitude & Phase

freqz(b,a)

% Impulse Response

figure

[h,n]=impz(b,a,50);

stem(n,h), grid on

title('Impulse Response')

% Step Response

figure

stepz(b,a)

title('Step Response')

% Group Delay

figure

grpdelay(b,a)

title('Group Delay')

% Pole-Zero Plot

figure
```

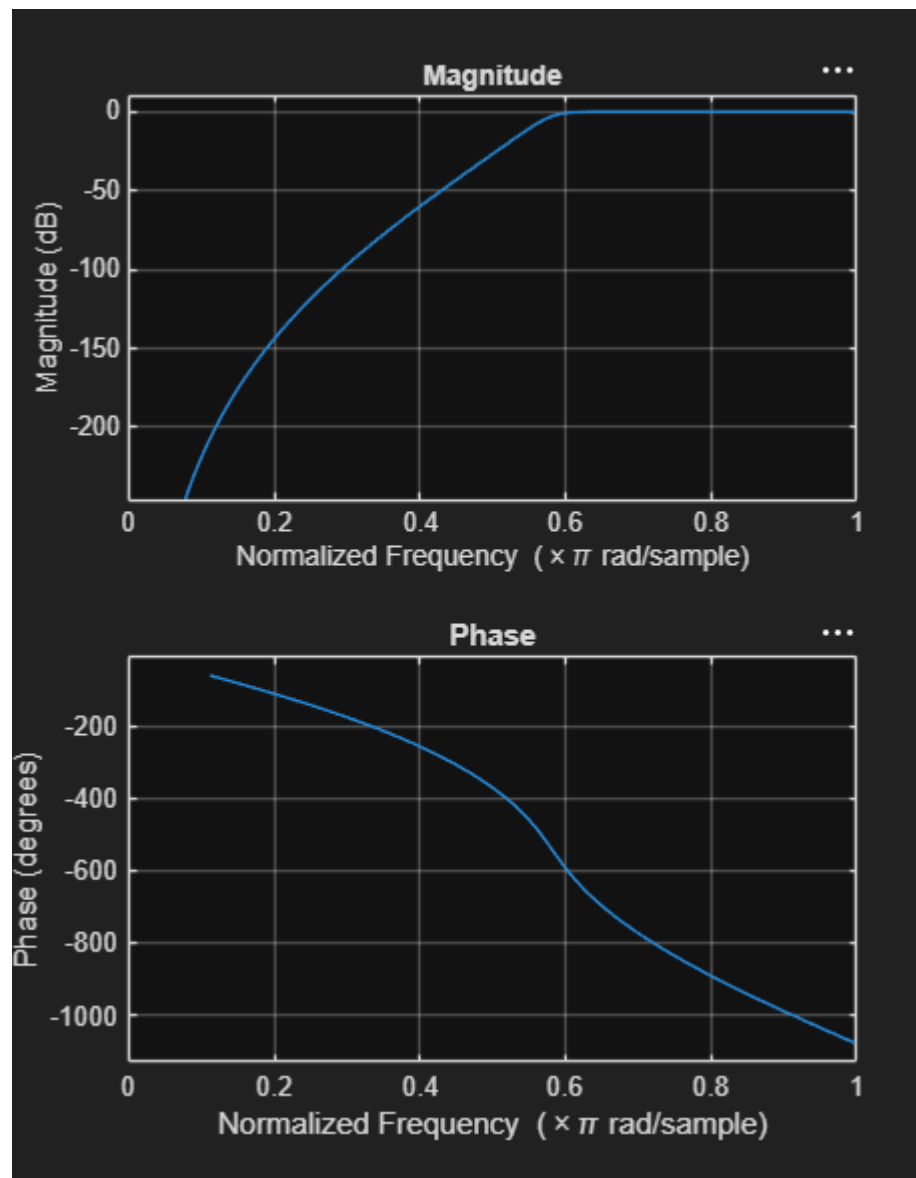
```
zplane(b,a)
```

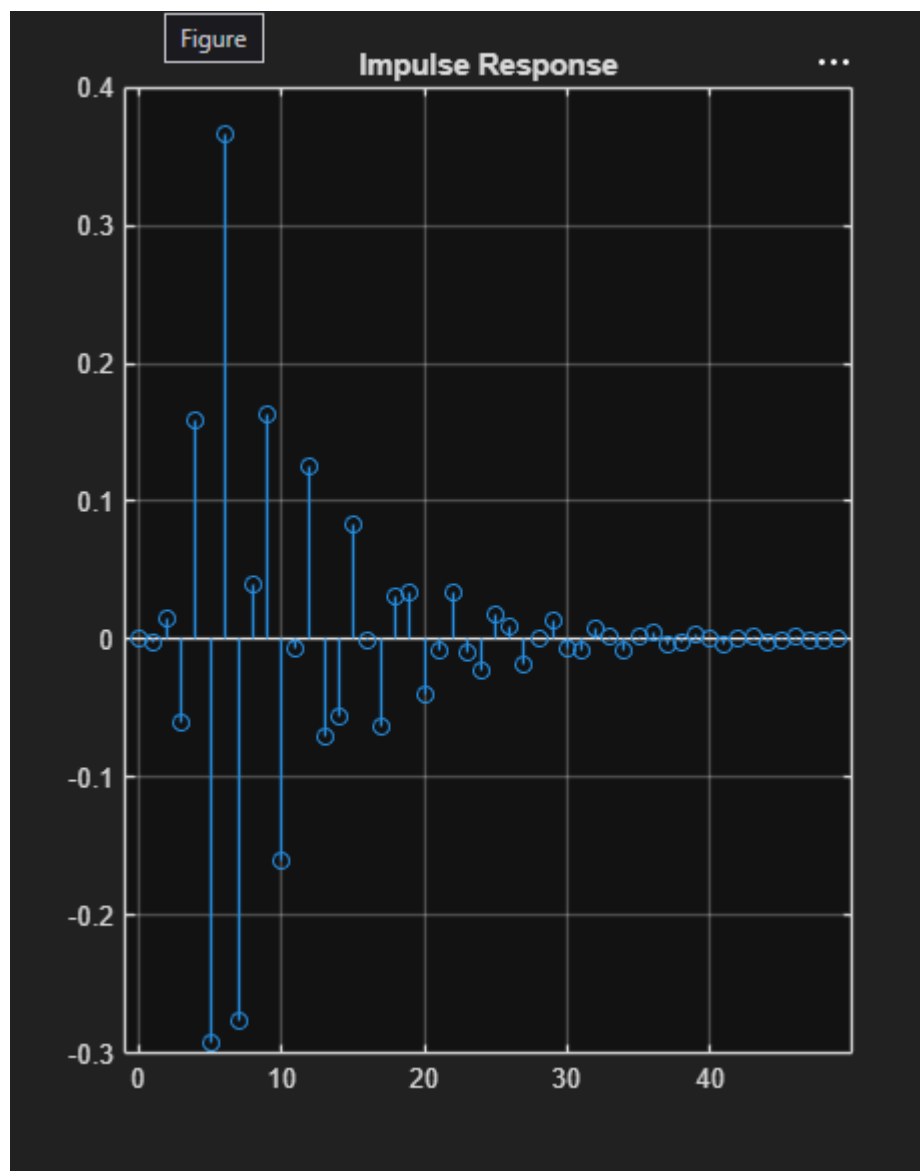
```
title('Pole-Zero Plot')
```

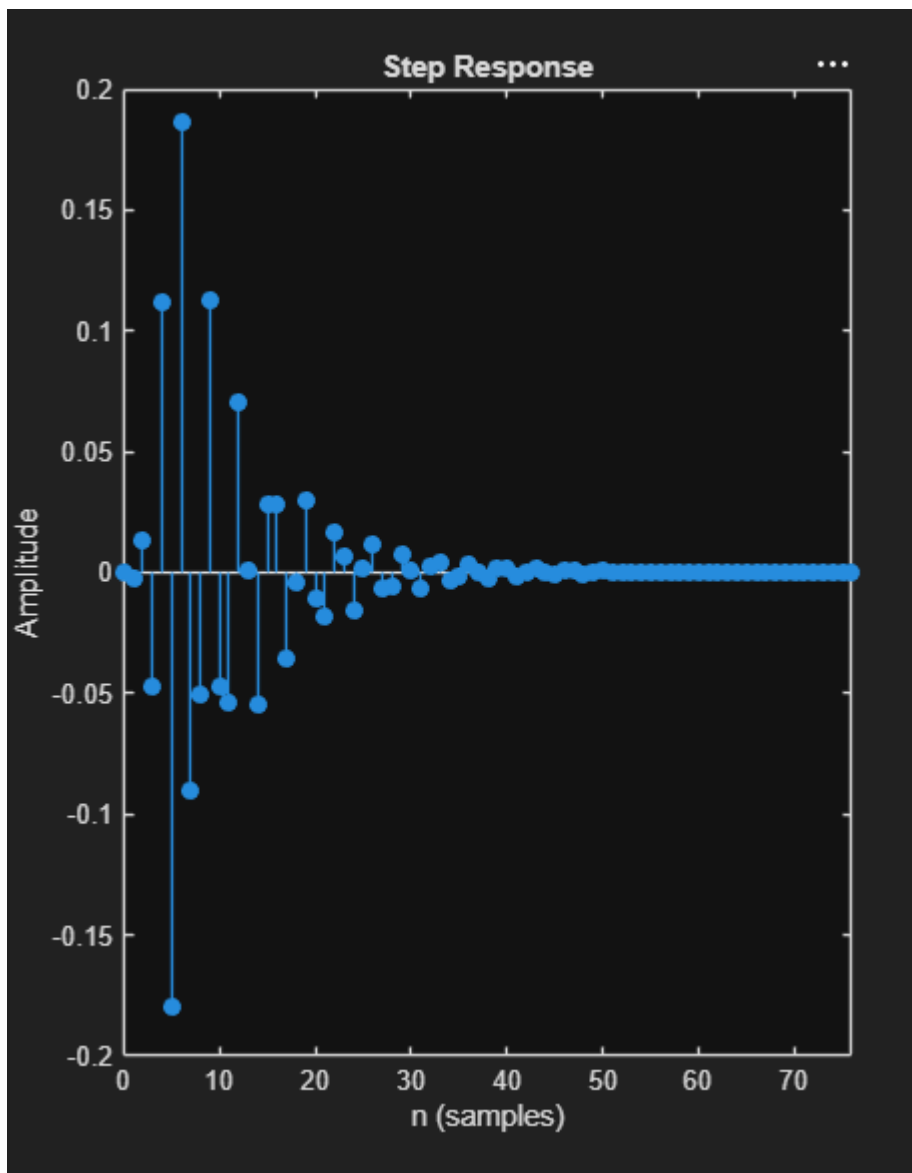
```
fprintf('Minimum Filter Order (N) = %d\n',N);
```

```
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

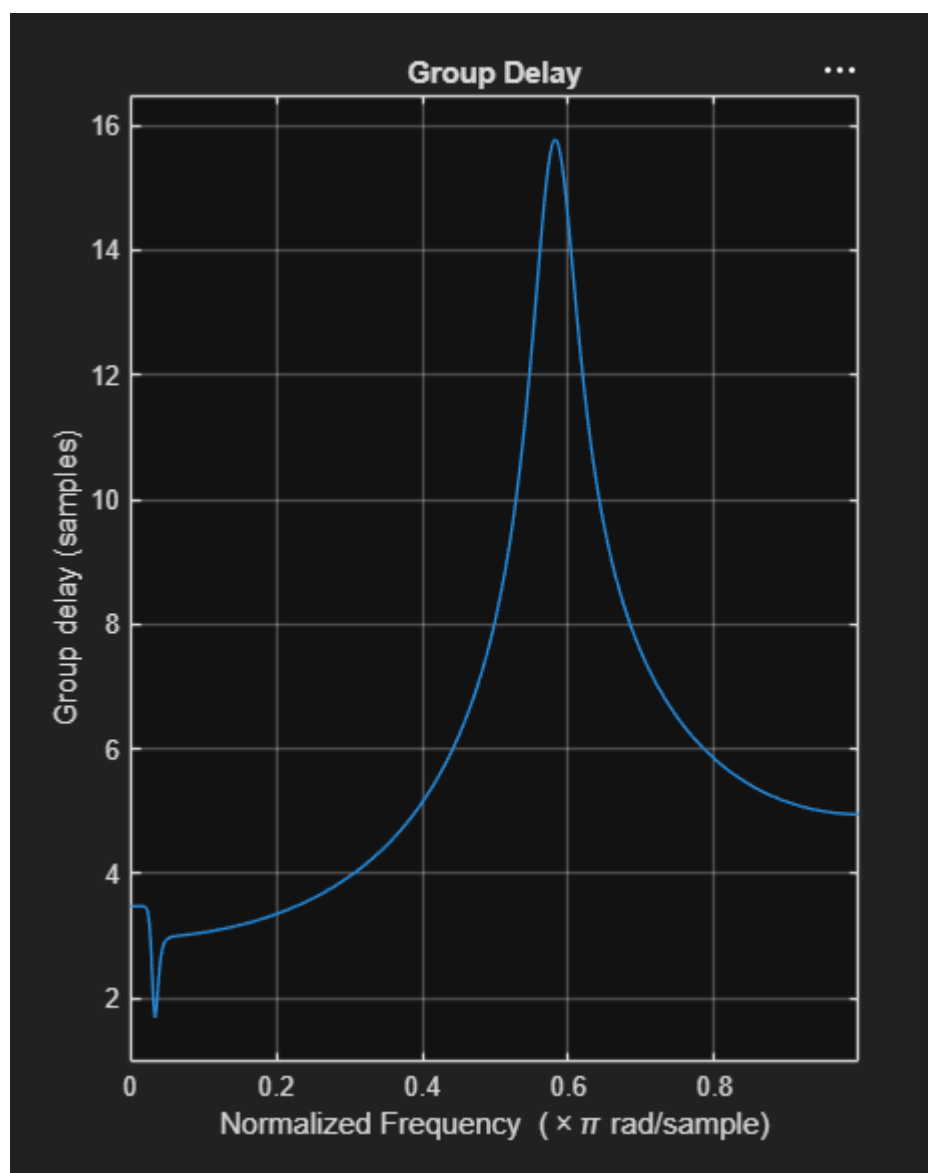
Output:

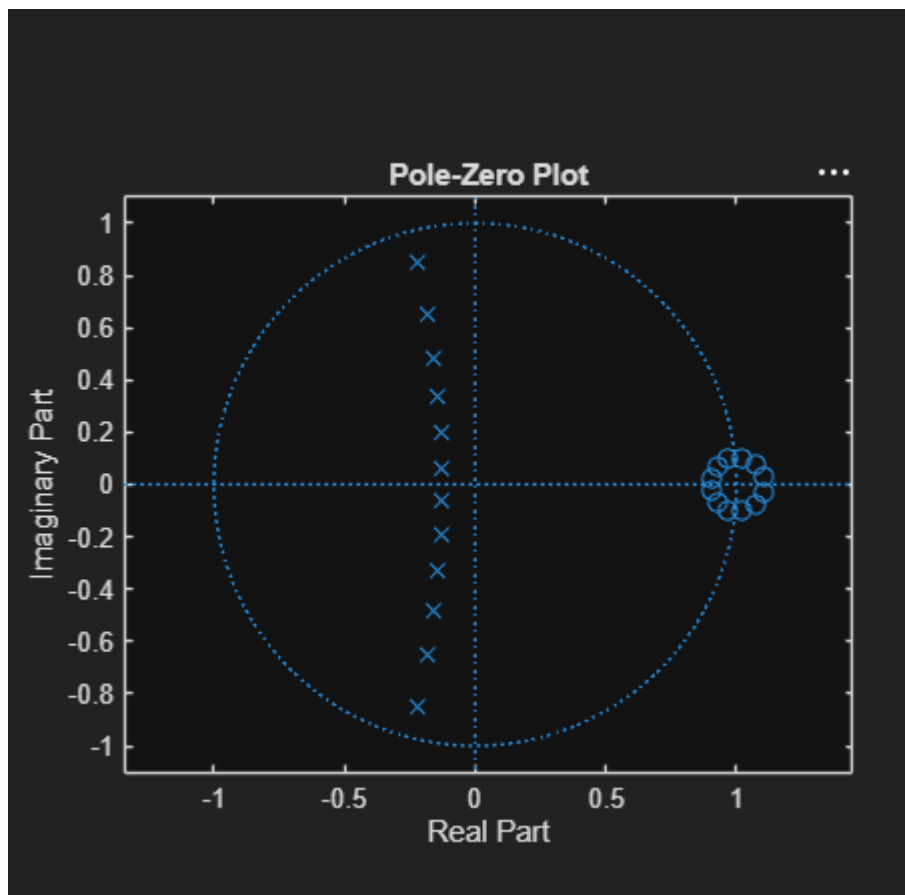












Minimum Filter Order (N) = 12  
Cutoff Frequency (Wn) = 0.5807  
>>